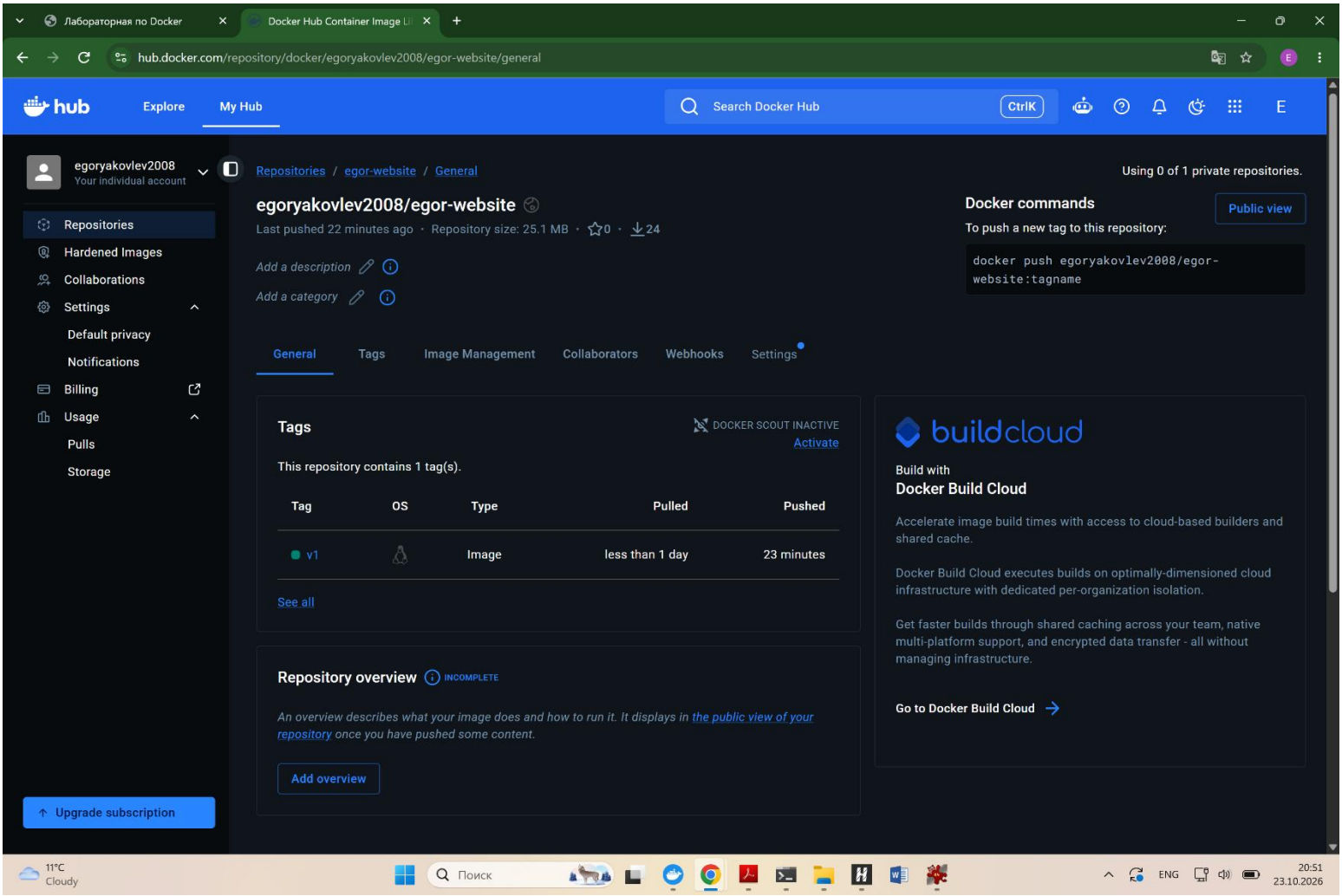
1.1) Собираем образ: docker build -t egor-wedsite:v1 .

```
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker build -t egor-website:v1 .
[+] Building 3.1s (8/8) FINISHED                                                
```

```
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker push egoryakovlev2008/egor-website:v1
The push refers to repository [docker.io/egoryakovlev2008/egor-website]
12b85cff06fd: Pushed
bfe72b1aeb47: Pushed
44136fa355b3: Pushed
7b229517ed22: Pushed
657023555a2a: Pushed
7cc330898115: Pushed
26185d08a3ec: Pushed
2ec5fb117ffa: Pushed
9b0f7dfe35b2: Pushed
b9b388a21ecf: Pushed
4732772eb688: Pushed
v1: digest: sha256:a7839071d27fb18c19acd079b9e6837b9f8b441514207c80b69dfa9873d9a5d1 size: 856
```

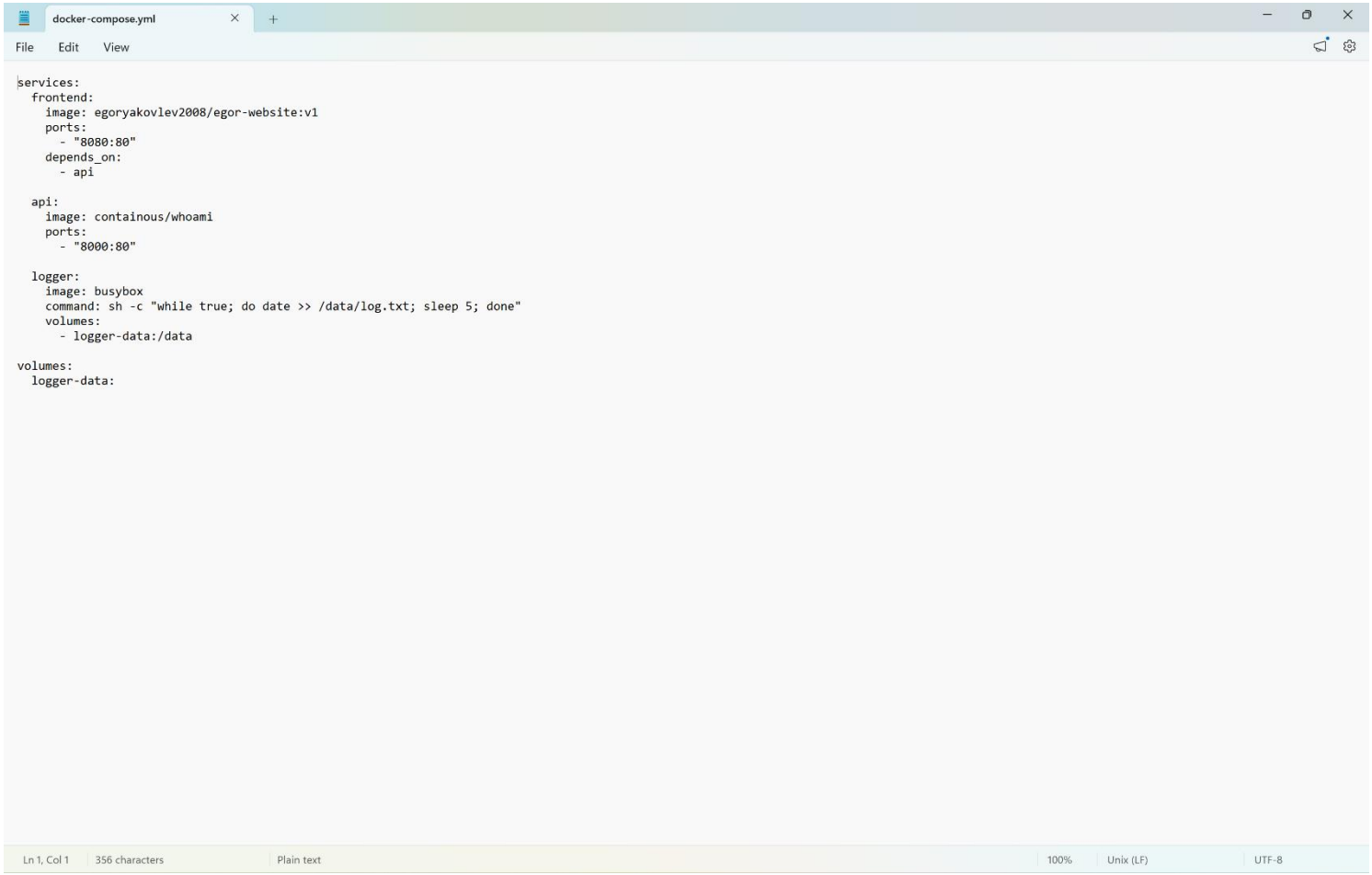
1.4) Проверка (скриншот из DockerHub):



1.5) Удаляем локальный образ и пуллим его обратно: docker rmi egoryakovlev2008/egor-website:v1 ; docker pull egoryakovlev2008/egor-website:v1

```
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker rmi egoryakovlev2008/egor-website:v1
Untagged: egoryakovlev2008/egor-website:v1
Deleted: sha256:a7839071d27fb18c19acd079b9e6837b9f8b441514207c80b69dfa9873d9a5d1
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker pull egoryakovlev2008/egor-website:v1
v1: Pulling from egoryakovlev2008/egor-website
4732772eb688: Pull complete
7cc330898115: Pull complete
12b85cff06fd: Pull complete
7b229517ed22: Pull complete
26185d08a3ec: Pull complete
657023555a2a: Pull complete
9b0f7dfe35b2: Pull complete
2ec5fb117ffa: Pull complete
b9b388a21ecf: Pull complete
44136fa355b3: Download complete
bfe72b1aeb47: Download complete
Digest: sha256:a7839071d27fb18c19acd079b9e6837b9f8b441514207c80b69dfa9873d9a5d1
Status: Downloaded newer image for egoryakovlev2008/egor-website:v1
docker.io/egoryakovlev2008/egor-website:v1
egoryak@DESKTOP-8IB7VLA:~/docker-lab$
```

2.1) Скриншот содержимого docker-compose.yml:



2.2) Запускаем и проверяем состояние: docker compose up –d ; docker compose ps

```
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker compose up -d
[+] up 12/12
 ✔Image busybox                  Pulled                                  7.3s
 ✔Image containous/whoami        Pulled                                  6.9s
 ✔Network docker-lab_default     Created                                 0.1s
 ✔Volume docker-lab_logger-data  Created                                 0.0s
 ✔Container docker-lab-logger-1  Started                                 1.3s
 ✔Container docker-lab-api-1     Started                                 0.7s
 ✔Container docker-lab-frontend-1 Started                                0.9s
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker compose ps
NAME                IMAGE                          COMMAND                  SERVICE    CREATED         STATUS         PORTS
docker-lab-api-1        containous/whoami                "/whoami"                api        3 minutes ago   Up 3 minutes   0.0.0.0:8000->80/tcp, [::]:8000->8
0/tcp
docker-lab-frontend-1   egoryakovlev2008/egor-website:v1   "/docker-entrypoint.…"   frontend   3 minutes ago   Up 3 minutes   0.0.0.0:8080->80/tcp, [::]:8080->8
0/tcp
docker-lab-logger-1     busybox                          "sh -c 'while true; …"   logger     3 minutes ago   Up 3 minutes
```

2.3) frontend обращается к api по имени сервиса:

```
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker compose exec frontend sh
/ # curl http://api
Hostname: 8c67504e8ca3
IP: 127.0.0.1
IP: ::1
IP: 172.18.0.2
RemoteAddr: 172.18.0.3:52406
GET / HTTP/1.1
Host: api
User-Agent: curl/8.22.0
Accept: */*
```

2.4) Проверяем наличие тома:

```
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker volume ls | grep logger
local     docker-lab_logger-data
```

2.5) Проверка, что logger записывает логи:

```
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker compose exec logger cat /data/log.txt
Sat Oct 24 06:40:21 UTC 2026
Sat Oct 24 06:40:26 UTC 2026
Sat Oct 24 06:40:31 UTC 2026
Sat Oct 24 06:40:36 UTC 2026
Sat Oct 24 06:40:41 UTC 2026
Sat Oct 24 06:40:46 UTC 2026
Sat Oct 24 06:40:51 UTC 2026
Sat Oct 24 06:40:56 UTC 2026
Sat Oct 24 06:41:01 UTC 2026
Sat Oct 24 06:41:06 UTC 2026
```

Последние строки:

```
Sat Oct 24 07:34:01 UTC 2026
Sat Oct 24 07:34:06 UTC 2026
Sat Oct 24 07:34:11 UTC 2026
Sat Oct 24 07:34:16 UTC 2026
Sat Oct 24 07:34:21 UTC 2026
Sat Oct 24 07:34:26 UTC 2026
Sat Oct 24 07:34:31 UTC 2026
egoryak@DESKTOP-8IB7VLA:~/docker-lab$
```

3.1) Проверка, что том переживает пересоздание контейнера: docker compose down ; docker compose up -d

```
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker compose down
[+] down 4/4
 ✔ Container docker-lab-logger-1   Removed                    3.5s
 ✔ Container docker-lab-frontend-1 Removed                    0.6s
 ✔ Container docker-lab-api-1      Removed                    0.4s
 ✔ Network docker-lab_default      Removed                    0.2s
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker compose -d
unknown shorthand flag: 'd' in -d
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker compose up -d
[+] up 4/4
 ✔ Network docker-lab_default      Created                    0.1s
 ✔ Container docker-lab-logger-1   Started                    0.6s
 ✔ Container docker-lab-api-1      Started                    0.9s
 ✔ Container docker-lab-frontend-1 Started                    1.1s
```

3.2) Данные пережили пересоздание контейнера: docker compose exec logger cat /data/log.txt

```
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker compose exec logger cat /data/log.txt
Sat Oct 24 06:40:21 UTC 2026
Sat Oct 24 06:40:26 UTC 2026
Sat Oct 24 06:40:31 UTC 2026
Sat Oct 24 06:40:36 UTC 2026
Sat Oct 24 06:40:41 UTC 2026
Sat Oct 24 06:40:46 UTC 2026
Sat Oct 24 06:40:51 UTC 2026
Sat Oct 24 06:40:56 UTC 2026
Sat Oct 24 06:41:01 UTC 2026
Sat Oct 24 06:41:06 UTC 2026
Sat Oct 24 06:41:11 UTC 2026
Sat Oct 24 06:41:16 UTC 2026
Sat Oct 24 06:41:21 UTC 2026
```

Последние строки:

```
Sat Oct 24 08:03:07 UTC 2026
Sat Oct 24 08:03:12 UTC 2026
Sat Oct 24 08:03:17 UTC 2026
Sat Oct 24 08:03:22 UTC 2026
Sat Oct 24 08:03:27 UTC 2026
Sat Oct 24 08:03:32 UTC 2026
Sat Oct 24 08:03:37 UTC 2026
Sat Oct 24 08:03:42 UTC 2026
Sat Oct 24 08:03:47 UTC 2026
```

3.3) –v удаляет том: docker compose down –v ; docker compose up –d ; docker compose exec logger sh -c "cat /data/log.txt"

```
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker compose down -v
[+] down 5/5
 ✔ Container docker-lab-frontend-1 Removed                    0.6s
 ✔ Container docker-lab-logger-1   Removed                    3.4s
 ✔ Container docker-lab-api-1      Removed                    0.4s
 ✔ Volume docker-lab_logger-data   Removed                    0.0s
 ✔ Network docker-lab_default      Removed                    0.3s
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker compose up -d
[+] up 5/5
 ✔ Network docker-lab_default      Created                    0.1s
 ✔ Volume docker-lab_logger-data   Created                    0.0s
 ✔ Container docker-lab-api-1      Started                    0.9s
 ✔ Container docker-lab-logger-1   Started                    0.6s
 ✔ Container docker-lab-frontend-1 Started                    1.1s
egoryak@DESKTOP-8IB7VLA:~/docker-lab$ docker compose exec logger sh -c "cat /d
ata/log.txt"
Sat Oct 24 08:11:37 UTC 2026
Sat Oct 24 08:11:42 UTC 2026
Sat Oct 24 08:11:47 UTC 2026
Sat Oct 24 08:11:52 UTC 2026
Sat Oct 24 08:11:57 UTC 2026
Sat Oct 24 08:12:02 UTC 2026
Sat Oct 24 08:12:07 UTC 2026
Sat Oct 24 08:12:12 UTC 2026
```